LIVE SESSION | EXPERT DEEP DIVE | SUBSCRIPTION

Security Best Practices for SAP Cloud Integration

Mar 13, 2026 | Various times | Online | 2 hours

Overview

In this live session, we will explore how to secure data and integrations on SAP Cloud Integration.

Learning objectives

After participating in this live session, you will understand how to secure and monitor data flows in SAP Cloud Integration.

Your current experience in this topic

Beginner | Intermediate | Advanced

Roles

Architect, Administrator

View schedule

Speaker

Jitesh Maharaj

jitesh.maharaj@tvpsap.com

➡️ LinkedIn: jmaharaj

Over 20 Years of working with SAP Products

- SAP NetWeaver and Business Technology Platform Consultant
- SAP Authorized Trainer
- SAP YPP Instructor (SAP Analytics Cloud)
- SAP Integration Specialist

Agenda

SAP Integration Suite

- Overview

SAP Cloud Integration

- Capabilities
- Monitoring

SAP Integration Suite Security

- Overview

SAP Integration Security Best Practices

- Deep Dive

SAP Integration Suite

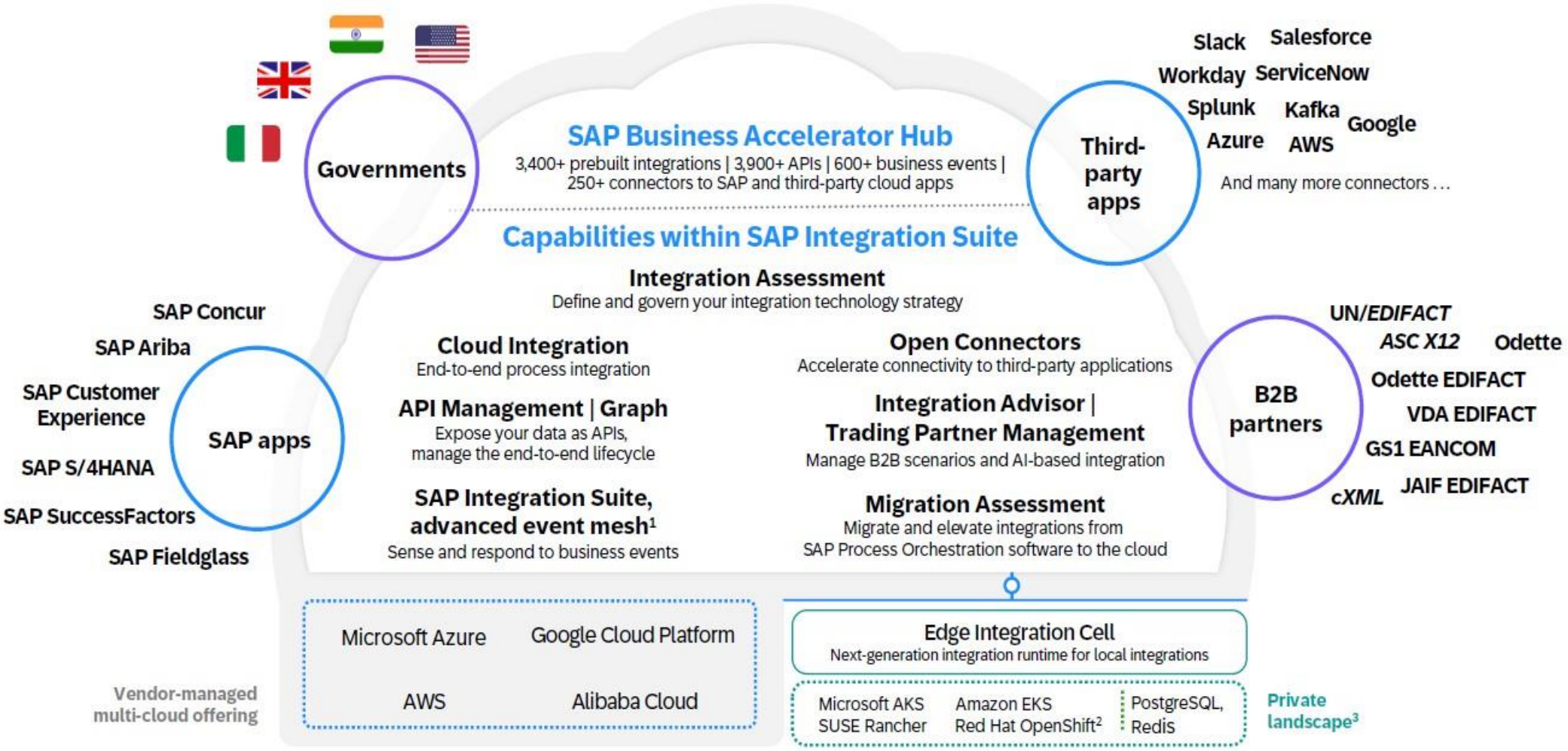
Public

SAP Integration Suite

Fully Maintained and Updated by SAP

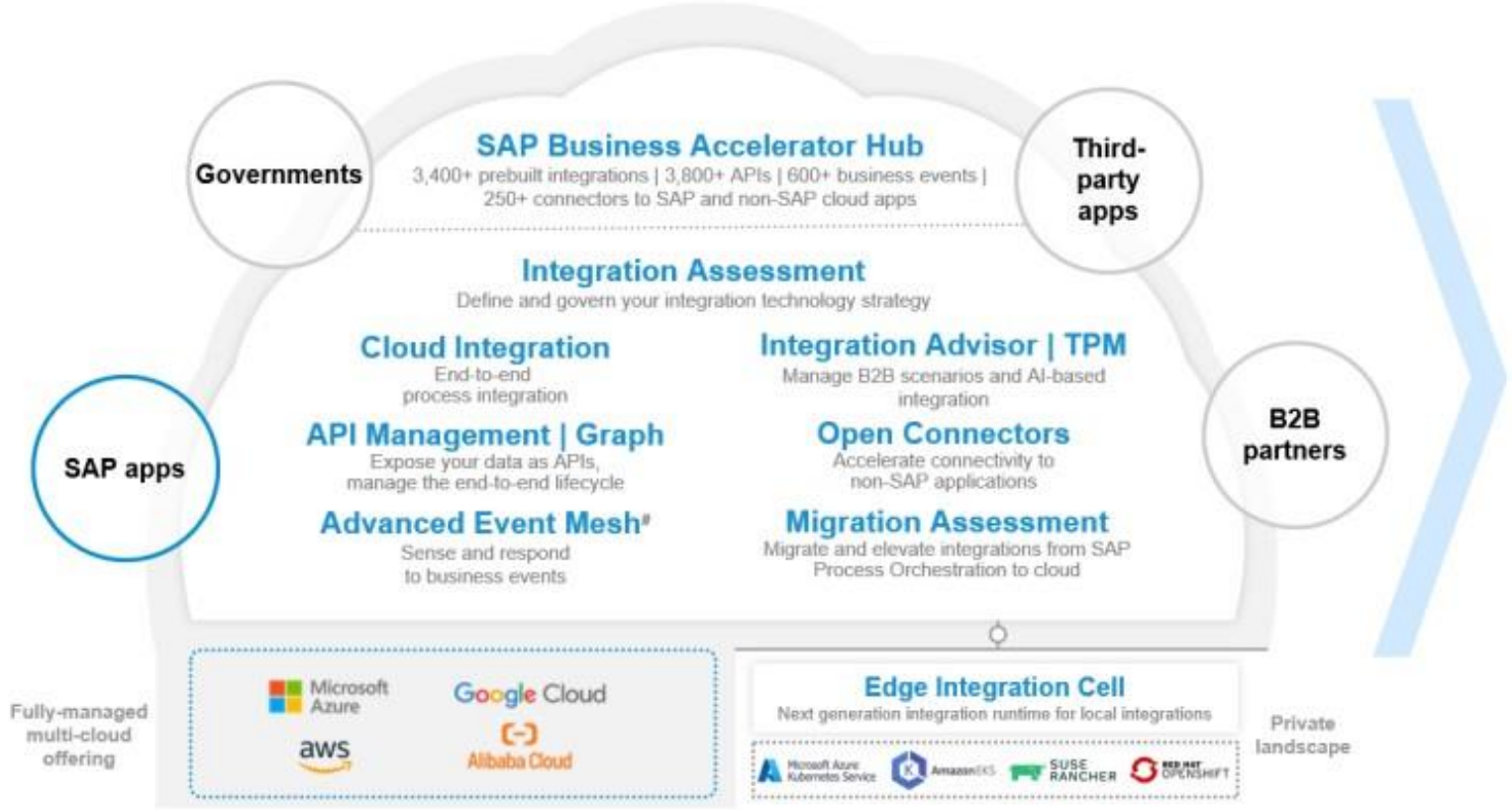
3400+ Prebuilt Integrations

1300+ Business Events



[1] Optional separate license | [2] Part of the road map, subject to change | [3] Private landscape corresponds to on premise or private cloud

Use cases



A2A integration
Chaining of business process steps between business applications by the exchange of transactional data

Master data integration
API-based exchange of master data across business applications

B2B integration
Chaining of business process steps between business partners by the exchange of transactional data leveraging standard B2B protocols

B2G integration
Transforming transactional data into predefined exchange formats and transfer it electronically to external legal or tax authority systems

API-based integration
Provisioning of omni-channel and secure access to business applications

Event-driven integration
Decoupling of applications, extensions or IoT data by messaging principles (pub/sub)

Digital integration hub
Large-scale, high-throughput APIs

Enterprise automation
Automation of business and IT processes with AI-powered tools

Integration in a nutshell

Process Integration

- Connect everything
- Automate and optimize processes
- Business Insights

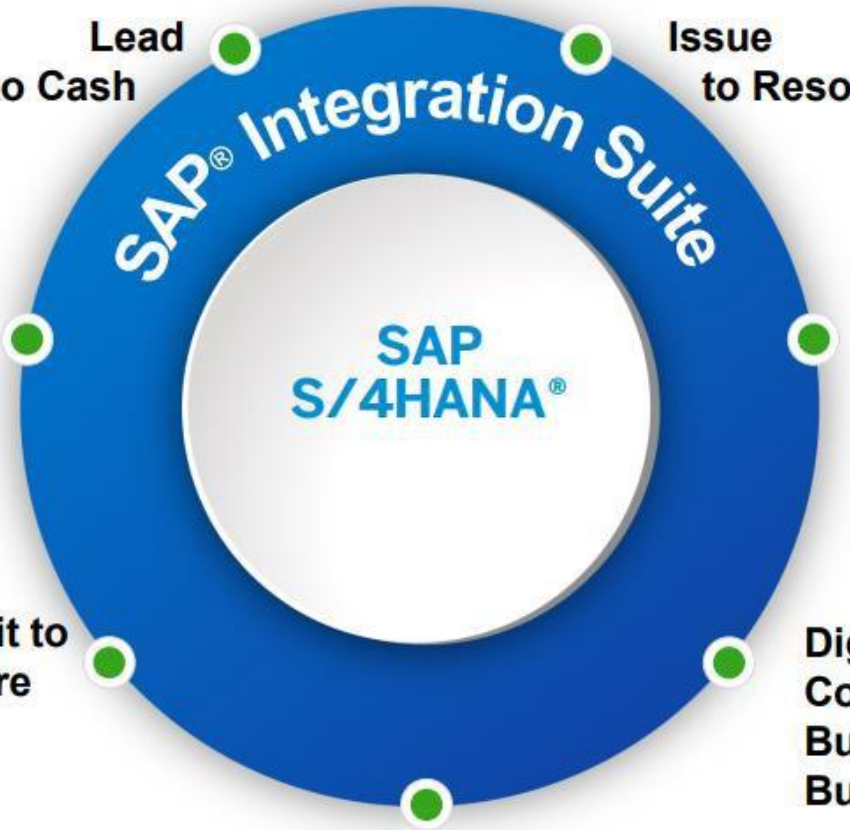
API-Led Integration

- Simplify Connectivity
- Usage and Analytics
- Break down silos

Event-Driven Integration

- Real-time data
- Decoupled Architecture
- Modular Adaptability

Seamlessly connect all your core business processes



Lead to Cash

- Lead management
- Segmentation and campaign management
- Omni-channel order management
- Trial to subscription

Issue to Resolution

- Issue detection to problem identification
- Develop solution to resolution
- Return to replace
- Closed loop feedback

Source to Pay

- Create or maintain suppliers
- Prepare purchase order
- Receive goods from supplier
- Enable payment

Payment Integration

- External payments system integration
- Treasury workstation payment integration

Recruit to Retire

- Payroll integration
- Benefits & compensation
- Recruiting & onboarding
- Time management
- Travel to expense

Digital Compliance, Business to Business (B2B)

- eDocuments
- eVAT/Invoice
- Payroll

Plan to Production

- Third-party warehouse integration
- Manufacturing Execution System integration

Fully Maintained and Updated by SAP

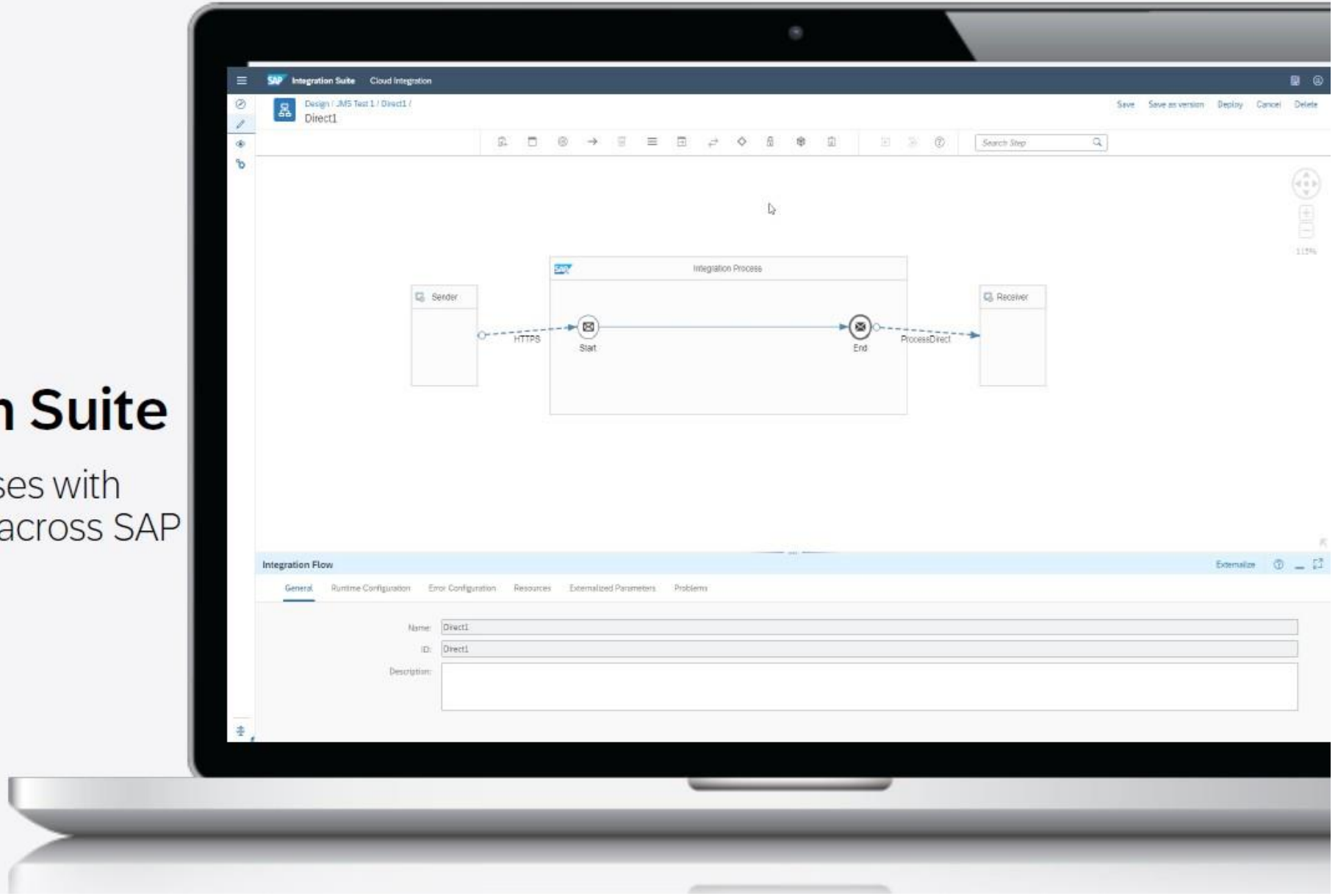
3400+ Prebuilt Integrations

1300+ Business Events

Public

SAP Integration Suite

Connect apps & processes with prebuilt tools & content across SAP & non-SAP.



Capabilities

Public

Capabilities of Integration Suite

Capabilities

Develop and manage enterprise-wide integration across heterogeneous landscapes.

To know how to add and activate capabilities, and assign capability-specific roles to users, see Capabilities.

Build Integration Scenarios

Discover, design, and operate scenarios for end-to-end process integration with Cloud Integration.

Discover Integrations •••

Manage APIs

Discover, design, and govern APIs for API consumers with API Management.

Discover APIs Configure APIs

Manage Trading Partners

Design and operate B2B scenarios with Trading Partner Management.

Manage Trading Partners

Implement Interfaces and Mappings

Design interfaces and mappings using crowdsourcing and machine learning with Integration Advisor.

Discover Type Systems •••

Access data in SAP Business Suite

Access business data in the SAP Business Suite with OData Provisioning.

Extend Non-SAP Connectivity

Connect to non-SAP cloud applications from your integration scenarios with Open Connectors.

Discover Connectors •••

Manage and Provide Integration Technology Guidance

Define, document, and govern your integration strategy powered by ISA-M with Integration Assessment.

Create Request

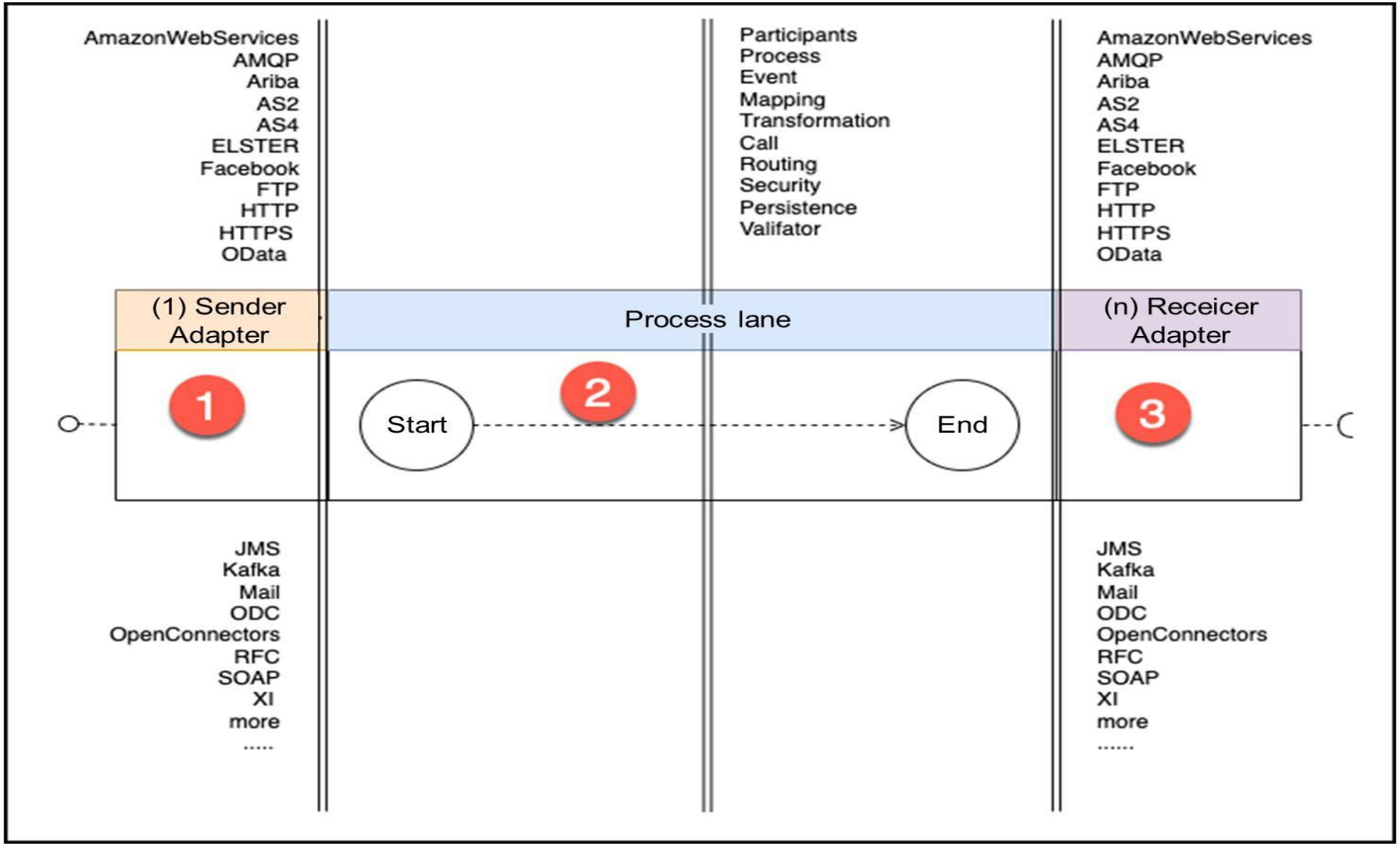
Assess Migration Scenarios

Estimate the migration effort for SAP Process Orchestration scenarios with Migration Assessment.

Create Requests

Manage Capabilities

Cloud Integration



Addon on Capability: Business Accelerator Hub

SAP Business Accelerator Hub | Integrations | Data Products | SAP Build | Analytics | Transformation | Explore | JM

Integration Flow Design Guidelines - Apply Highest Security Standards

Add To WorkSpace | Mark as Favorite | Subscribe

This integration package contains integration flows to illustrate the design guidelines for applying highest security standards.

Overview | Integration Flow | Documents

Sort by: Updated

Find

Integration Flow

Generic Receiver

Integration flow mocking a receiver system for test purposes

Version 1.0.1

Integration Flow

Apply Security - Upload WSDLs as Integration Flow Resource

This Integration Flow shows how you can upload WSDL and XSD files as resources to avoid relying on external references

Version 1.0.0

Integration Flow

Apply Security - Use CSRF Protection - Sender Channel

This Integration Flow shows how you ensure CSRF protection for an HTTP sender channel

Version 1.0.0

Integration Flow

Apply Security - Use CSRF Protection - Receiver Channel

This Integration Flow shows how you ensure CSRF protection for an HTTP receiver channel

Version 1.0.2

Integration Flow

Apply Security - Apply Message

Integration Flow

Apply Security - Remove Sensitive

Integration Flow

Apply Security - Apply Message

Integration Flow

Apply Security - Use Client

API Security: Policy Templates

SAP Business Accelerator Hub | Integrations | Data Products | SAP Build | Analytics | Transformation | Explore | JM

APIs

Find APIs to integrate and extend.

Packages | All | SOAP | ODATA V2 | ODATA V4 | Delta Sharing | GraphQL | REST | Policy Template

Showing 1-12 of 25 results

< Previous | 1 | of 3 | Next > | Sort by: Updated

Find

Policy Template

Developer Hub Governance with SAP Build Process Automation

This policy template facilitates the transformation of the Developer Hub Service Provider Interface (SPI) when implementing SAP Build Process Automation to establish your governance workflow

Version 1 ACTIVE

Policy Template

SuccessFactors Principal Propagation via Entra ID

This policy template bundle enables citizen developers across the SAP and Microsoft ecosystem to leverage single sign on (SSO) / principal propagation into SAP SuccessFactors with Microsoft Entra ID. Thi...

Version 1

Policy Template

SuccessFactors_OAuth2SAMLAssertion

This policy template facilitates easy consumption of SAP SuccessFactors APIs protected by OAuth

Version 1

Policy Template

JSON Threat Protection

This policy template assists you in securing your APIs from JSON threats for APIs managed by API Management.

Version 1

Policy Template

Principal Propagation via Entra Id

This feature rich policy template enables citizen developers across the SAP and Microsoft ecosystem solve the problem of single sign on / principal propagation into SAP systems with Microsoft Entra ID as the...

Policy Template

Principal Propagation via SAML

This policy template assists you in SAML User propagation from the application to back-end system via SAP API Management

Policy Template

JWT Token Verification Policy template

This policy template assists to generate and verify JWT token for APIs managed by API Management.

Policy Template

XML Threat Protection

This policy template assists you in securing your APIs from XML threats for APIs managed by API Management

Sample Monitoring Logs

Integration Suite

Overview / Monitor Message Processing

Message Status Overview | Hide Filter Bar

Time: Past Hour | Status: Completed | Type: All | Package: All | Artifacts: Apply Security - Remove Sensit... | or | ID: AGmzdidumOtP9qsk-AcqmZxwptsC

Messages (1) 1 / 1

Artifact Name	Status
Apply Security - Remove Sensitive Content Mar 13, 2026, 04:27:51	Completed 92 ms

Apply Security - Remove Sensitive Content

Last Updated at: Mar 13, 2026, 04:27:51

Status | Properties | Logs | Artifact Details

Status

Message processing completed successfully.

Processing Time: 92 ms

Properties

Message ID: AGmzdidumOtP9qsk-AcqmZxwptsC

Correlation ID: AGmzdid2gYueMVe4Q-wLLfshue48

Retention Periods

Logs

Open Text View

Trace data is removed after the configured retention time, typically 1 hour.

Log Level: Trace

Instance ID: 0

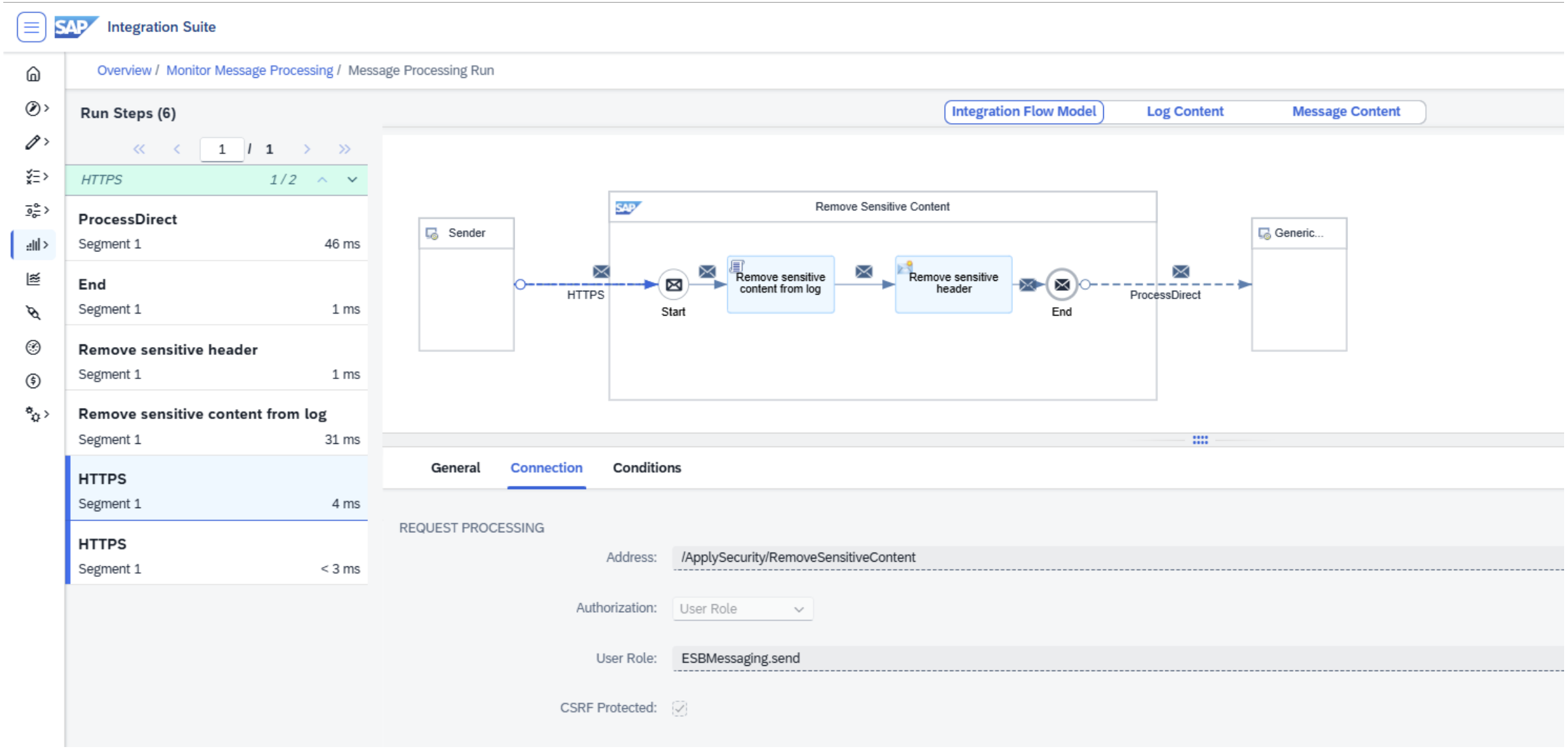
Artifact Details

Manage Integration Content

View deployed Artifact

Navigate to Artifact Editor

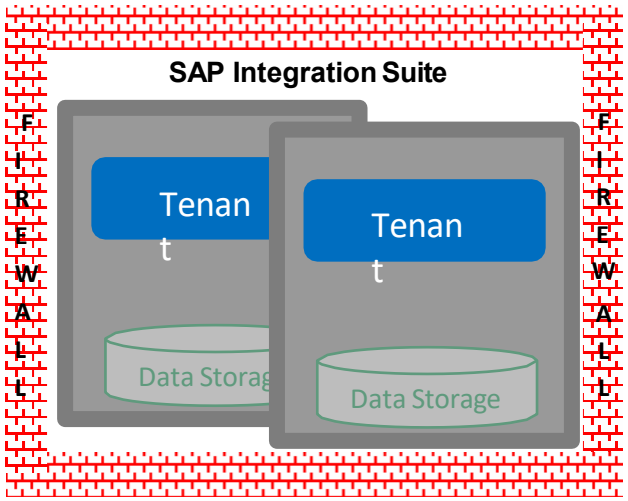
Log Level Set to Trace

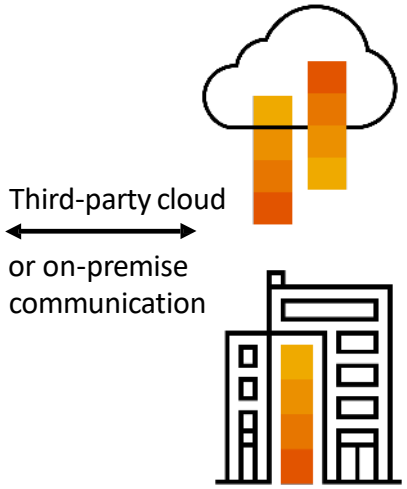


Security

Public

Security aspect of Cloud Integration Suite





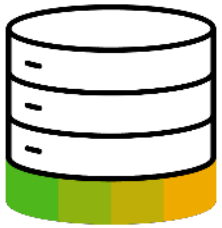
Secure Certified Data Centers

- Compliance standards: **ISO 22301, ISO 27001, SOC 1, SOC 2, C5, CSA STAR, TISAX**
- Hundreds of surveillance cameras with digital recording
- Fully monitored doors
- Tens of thousands of environmental sensors
- Security and facility support team onsite 24x7
- Biometric access to secured areas
- Multiple redundant internet connections from multiple carriers
- Redundant data storage in two data center locations

➔ How Secure is the SAP Data Center?

Secure Provisioning and Infrastructure

- Multi-tenancy
- Data isolation
- Audit logs
- Tracing
- Roles based access
- Data privacy and protection
- Data storage and backup
- Malware scanning of uploaded resources



Secure Communication

- Transport level: HTTPS, SFTP, FTPS, TLS, SMTP, POP3, IMAP
- Payload level: Encryption, decryption, signing, signature verification:
  - PGP, WS security, XML digital signature, PKCS#7
- Authentication:
  - OAuth 2.0, X.509 certificates, PGP keys, SAML, JWT, known host file, basic authentication
- Secure storage of key stores in data base
- Encrypted data persistency

Strong focus on Security
Cloud Integration ensures Cloud Security at multiple levels

- Tenant isolation to separate data

- Data flow security on transport level
  - HTTP over TLS
  - HTTP authentication using an X.509 client certificate
  - SSH File Transfer Protocol (SFTP)
  - SMTP, POP3, and IMAP
  - Authentications such as JWT, SAML, X509, Basic

- Data flow security on message level
  - Digital encryption and signatures

- Data storage security
- Audit logging and tracing (stored for a defined retention time)
- Physical data storage
- User management and authorization
  – Role based authorization concept
- Access management
  – Client certificate authentication
  – Single sign-on
- Malware scanning of uploaded resources

Guidelines for Security and Access Management Aspects

- Integration developer has to make sure that the integration flows are designed in a way that ensures that the highest security standards are applied. Refer to Concepts of Secure Communication and Apply the Highest Security Standards guidelines.
- Define roles and permissions for each Persona and for each environment. You can make use of the predefined authorization groups or tailor the permissions to your own requirements by applying roles that are defined for individual tasks: Tasks and Permissions.
- It is **not** recommended to use basic authentication for inbound or outbound communication over HTTPS in productive scenarios. Instead, make use of client certificate or OAuth authentication. More info: HTTPS-Based Communication.
- Make use of Auditing and Logging Information if you suspect a security violation.
- Use API Management to apply additional security policies (URL masking, IP access control, validation/threat protection policies, etc.) and expose Cloud Integration endpoints to Developer Hub (formerly known as API Business Hub Enterprise) where they can be securily consumed by internal/external developers. More info: Streamline Exposing and Managing Connections to Your Cloud Integration Flows with API Management.
- Define Access Policies when you need to restrict the access to integration artifacts and their associated data to specific user group. More info: Cloud Integration – Access Policies: Defining Roles on Artifact Level.

Persisting and securing messages

Message level security: encryption

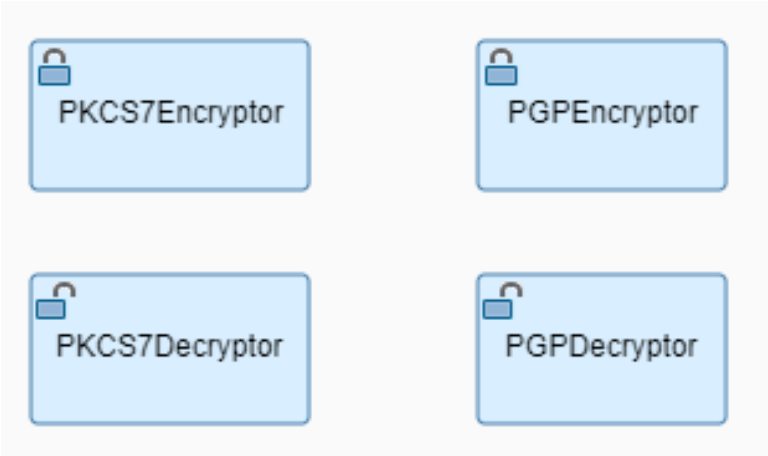
Encryption

- Cloud Integration encrypts messages with the public key/certificate of the receiver (stored in the Cloud Integration keystore)

- Standalone encryption/decryption
  - PKCS#7
  - PGP
- Adapter-based encryption/decryption
  - S/MIME: Mail adapter
  - WebService Security: SOAP, AS4

Decryption

- Cloud Integration decrypts messages with the own private/secret key (stored in the Cloud Integration keystore)
- In case of a key change inside Cloud Integration, all communication partners using encryption must be updated with the new public key



Persisting and securing messages

Message level security: signing

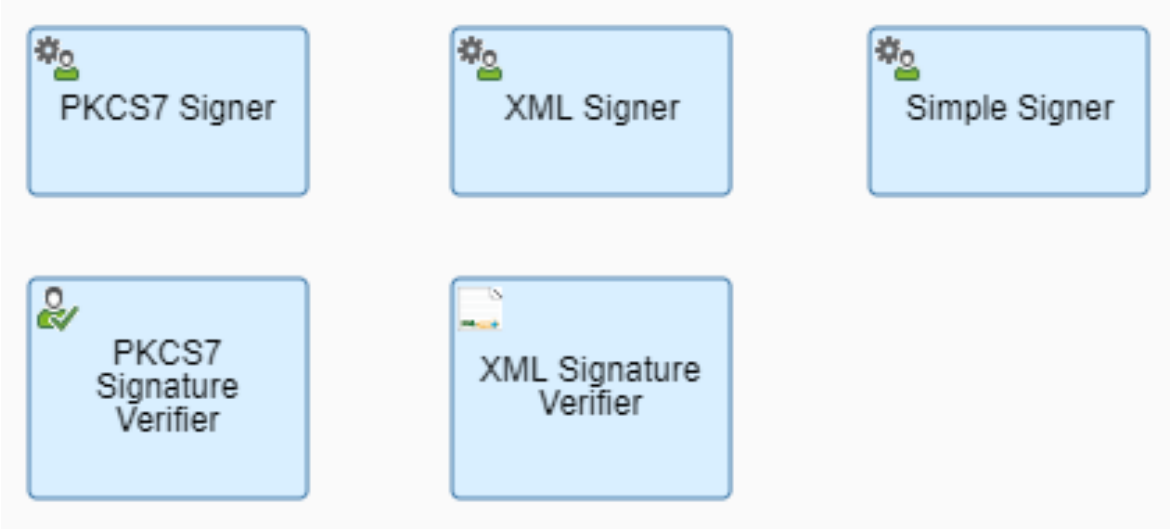
Signature

- Senders sign messages in order to prove that a message was sent by them
- Signing on Cloud Integration is done with the private/secret key (stored in the Cloud Integration keystore)
- Each tenant is provisioned with a key pair; more key pairs can be generated

- Standalone signature
  - PKCS#7
  - XML signature
  - Simple signature
- Adapter-based signature
  - S/MIME: Mail adapter
  - WS-Security: SOAP, AS4

Verification

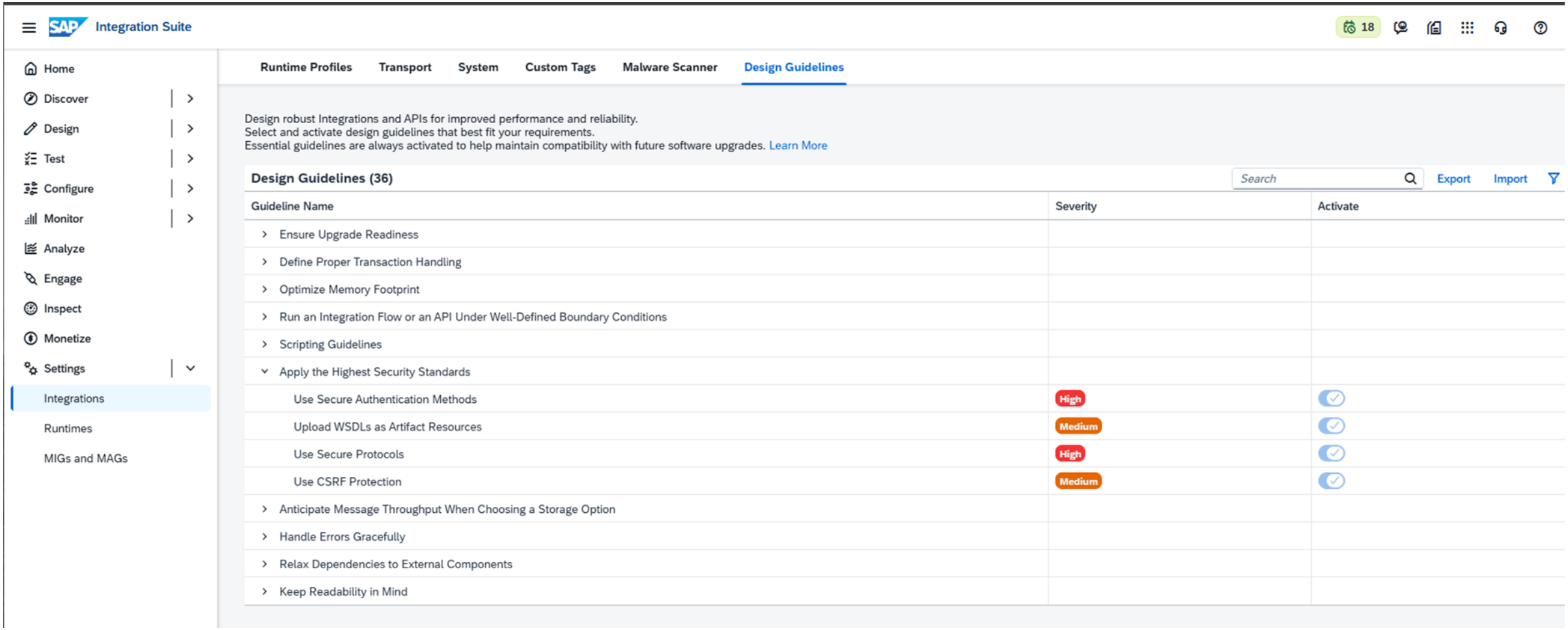
- Receiver verifies signatures to check if payload can be trusted
- Signatures are verified by Cloud Integration with the public key/certificate of the signer (stored in the Cloud Integration keystore)



Best Practices Security

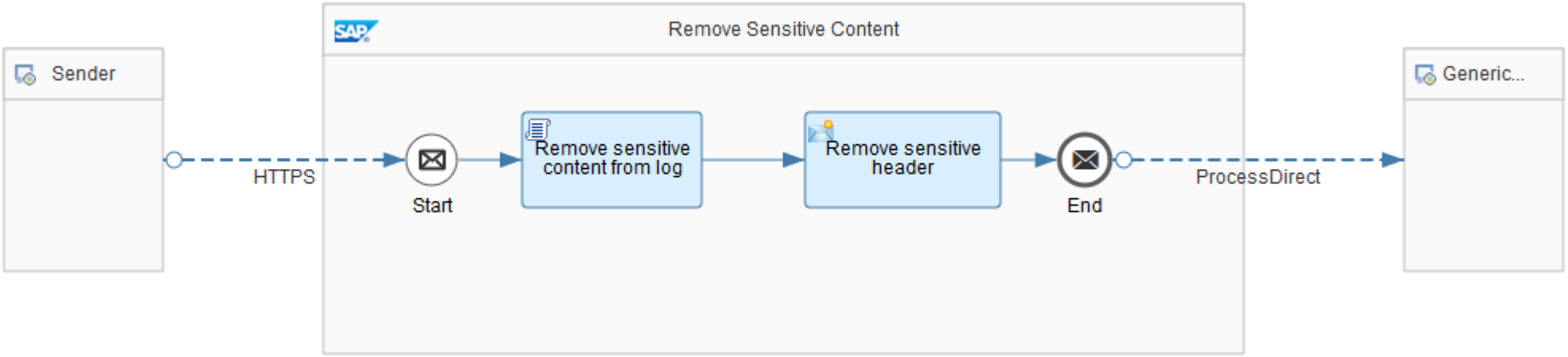
Public

Integration Design Guidelines



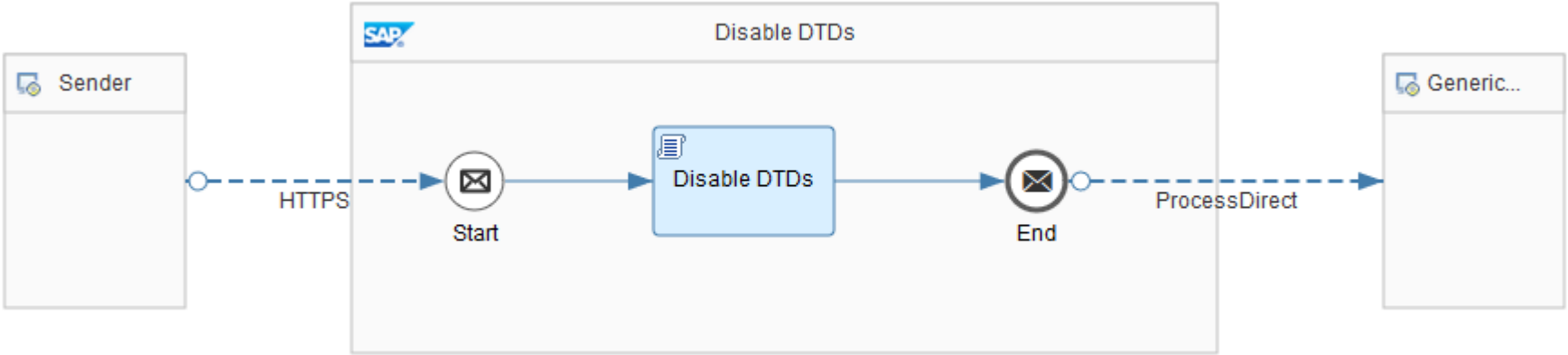
Remove Sensitive or Secret Content Before Logging or Forwarding

For general considerations on logging from a security point of view, see
https://cheatsheetseries.owasp.org/cheatsheets/Logging_Cheat_Sheet.html



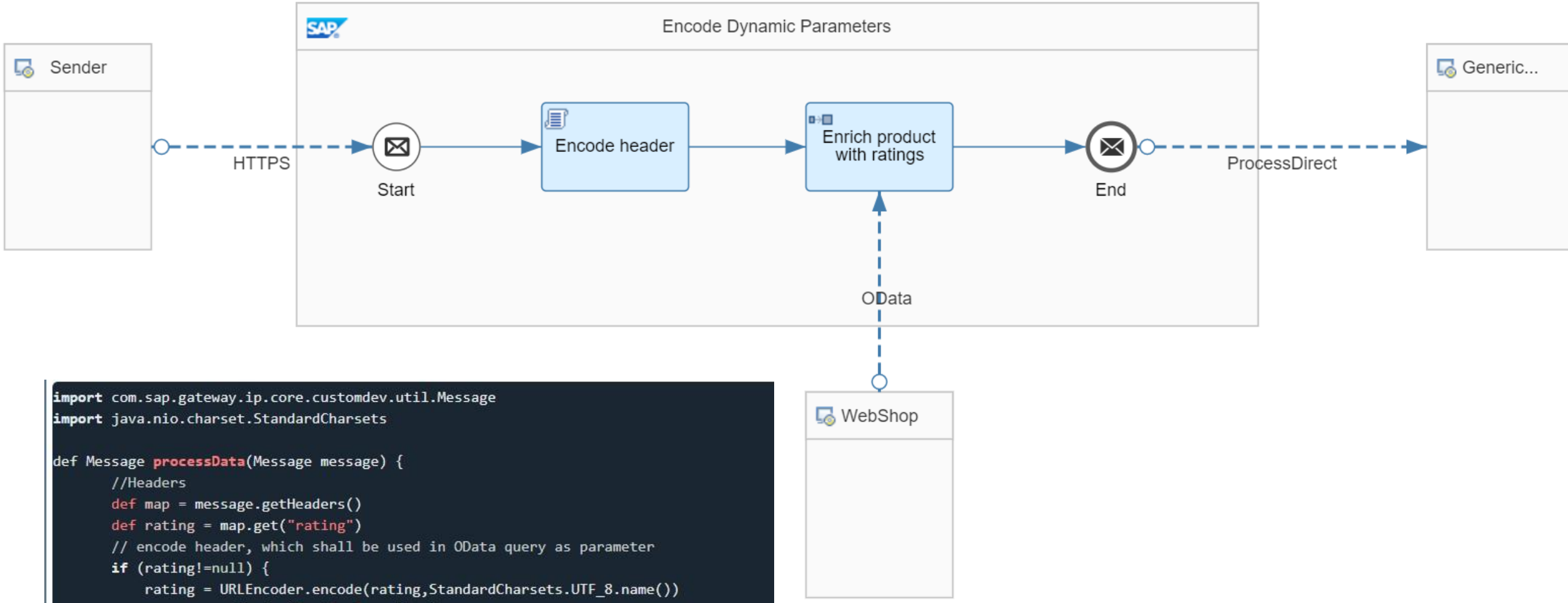

```
import com.sap.gateway.ip.core.customdev.util.Message;
import java.util.HashMap;
def Message processData(Message message) {
        def messageLog = messageLogFactory.getMessageLog(message)
        // Log Headers
        def map = message.getHeaders();
        // Log only needed headers (allow list)
        messageLog.setStringProperty("Content-Type (allowlist)",map.get("Content-Type"))
        messageLog.setStringProperty("Accept (allowlist)",map.get("Accept"))
        // exclude sensitive headers (block list)
        map.each{ k, v -> if (!k.equals("authorization"))  messageLog.setStringProperty(k+" (blocklist)", v) }
        return message;
}
```

Switch Off Resolving of External Entities




```
import com.sap.gateway.ip.core.customdev.util.Message;
import javax.xml.parsers.DocumentBuilderFactory;
import javax.xml.parsers.DocumentBuilder;
import javax.xml.parsers.ParserConfigurationException;
import org.xml.sax.InputSource;
import java.io.StringReader;
def Message processData(Message message) {
    def dbf = DocumentBuilderFactory.newInstance();
    // disallow the processing of doctype declarations
    def FEATURE = "http://apache.org/xml/features/disallow-doctype-decl";
    dbf.setFeature(FEATURE, true);
    def db = dbf.newDocumentBuilder();
    def is = new InputSource(message.getBody());
    def doc = db.parse(is);
    message.setBody(doc);
    return message;
}
```

Encode Dynamic Parameters



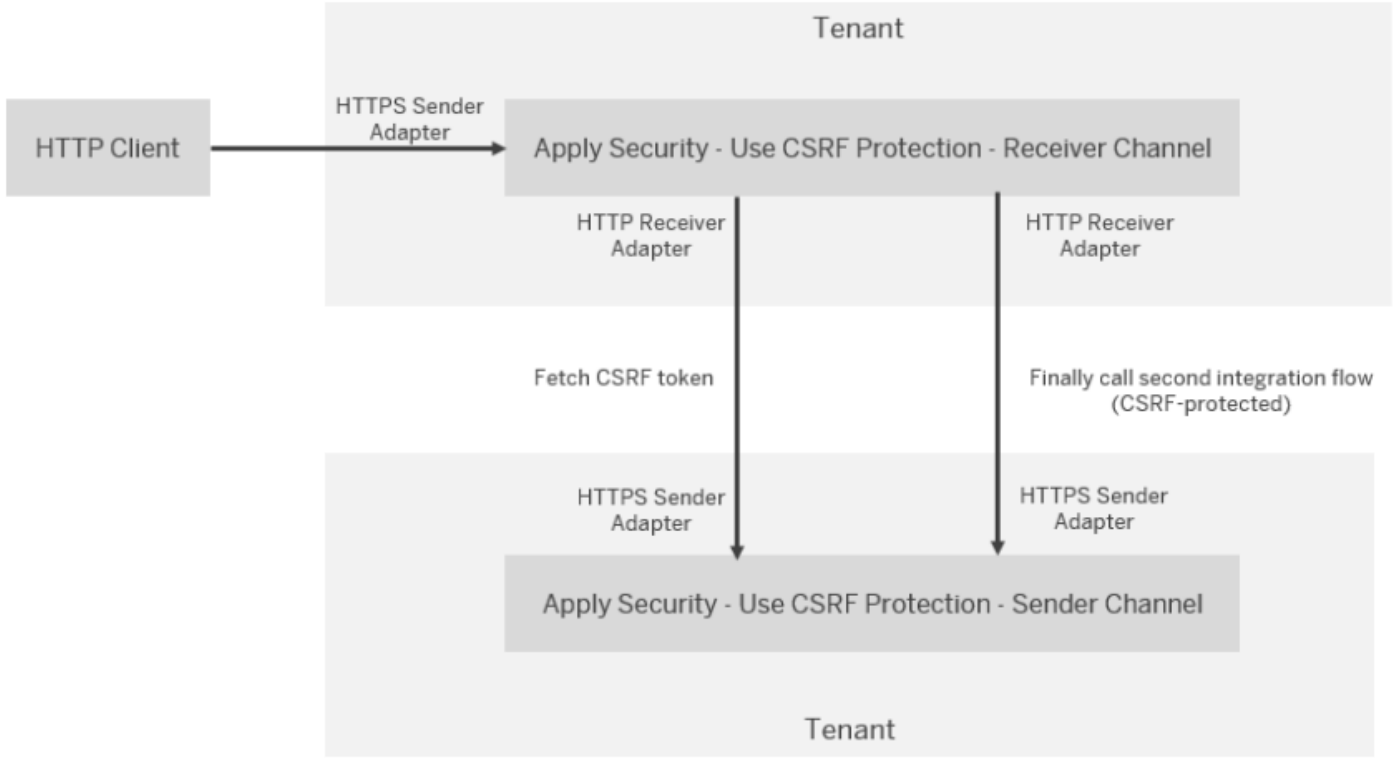

```
import com.sap.gateway.ip.core.customdev.util.Message
import java.nio.charset.StandardCharsets

def Message processData(Message message) {
        //Headers
        def map = message.getHeaders()
        def rating = map.get("rating")
        // encode header, which shall be used in OData query as parameter
        if (rating!=null) {
            rating = URLEncoder.encode(rating,StandardCharsets.UTF_8.name())
            message.setHeader("rating", rating)
        }
        return message
}
```

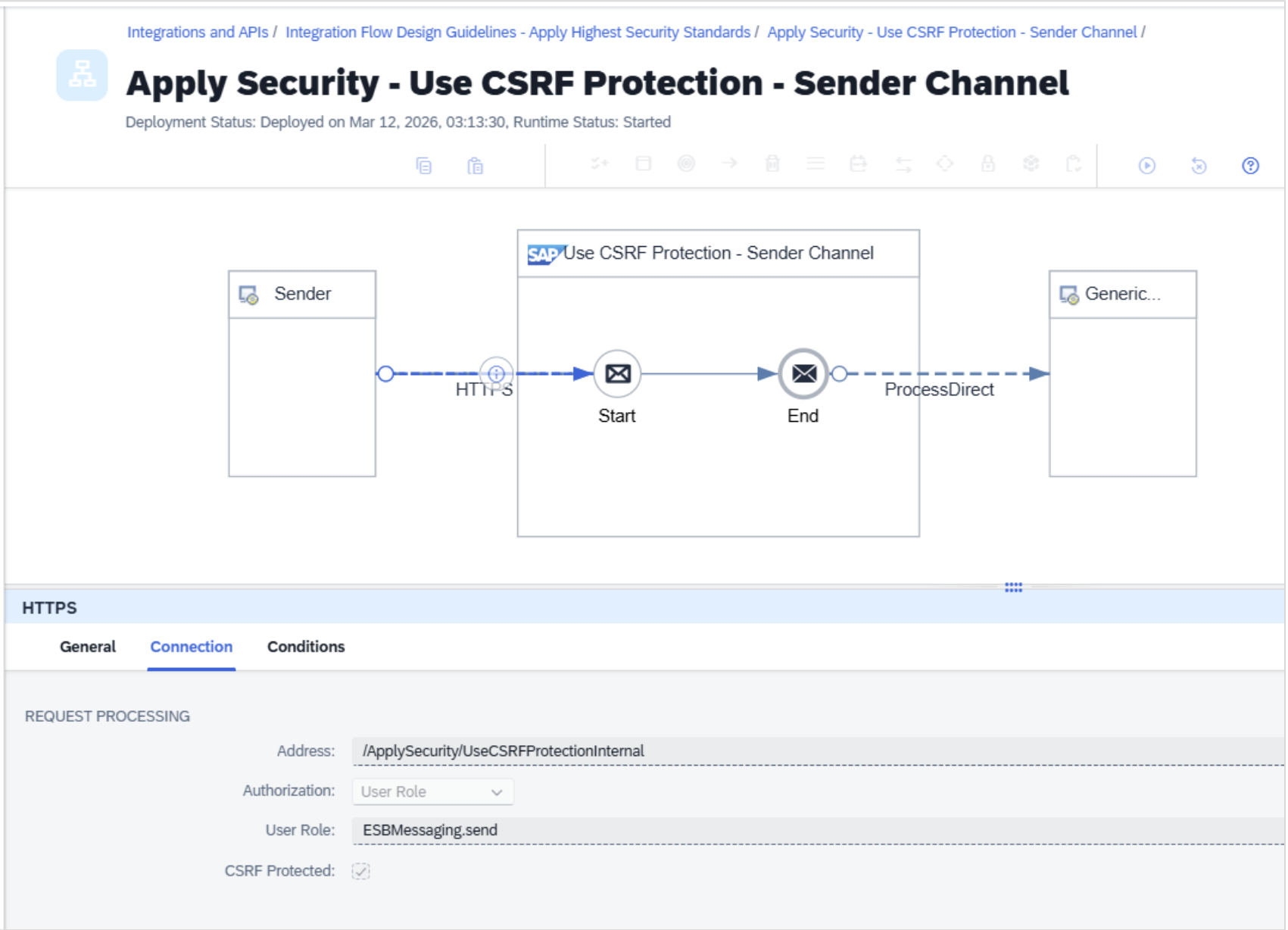
Use CSRF Protection - Cross Site Request Forgery

- **Apply Security - Use CSRF Protection - Receiver Channel**
- **Apply Security - Use CSRF Protection - Sender Channel**

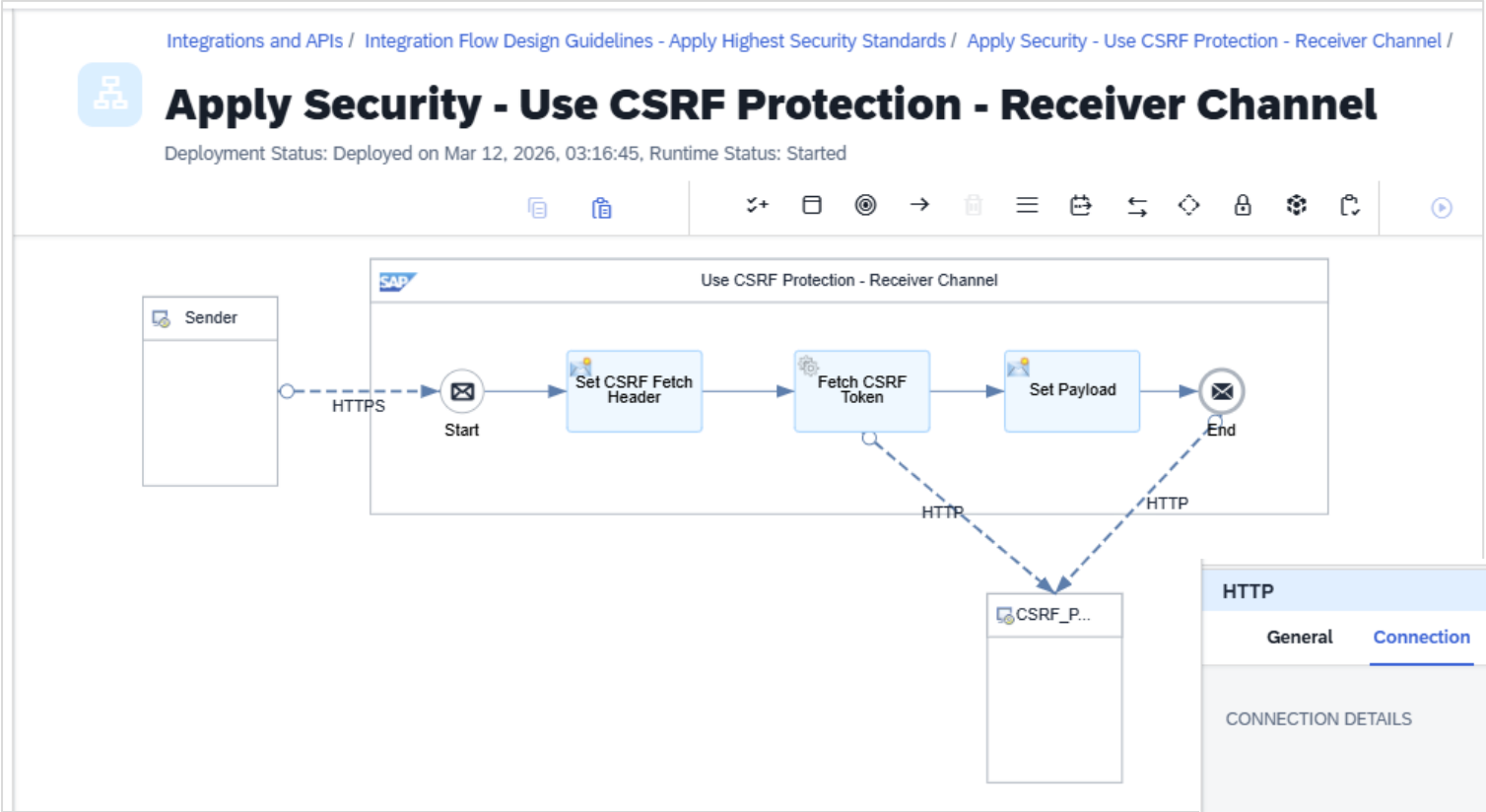
Both integration flows communicate with each other as shown in the figure.

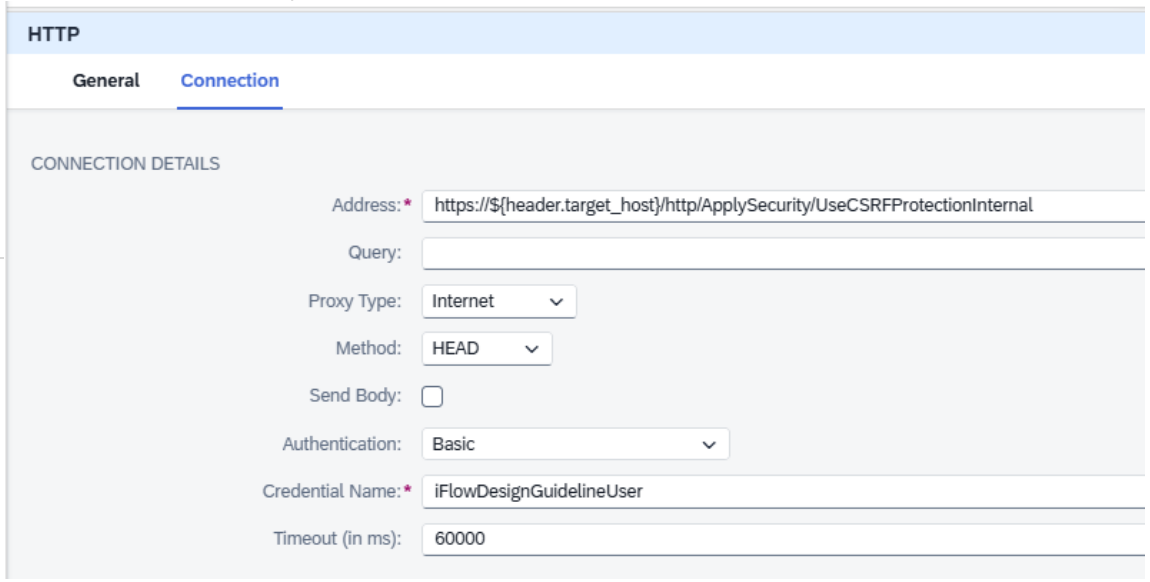


Implementation Steps for CSRF Protection – Sender Channel

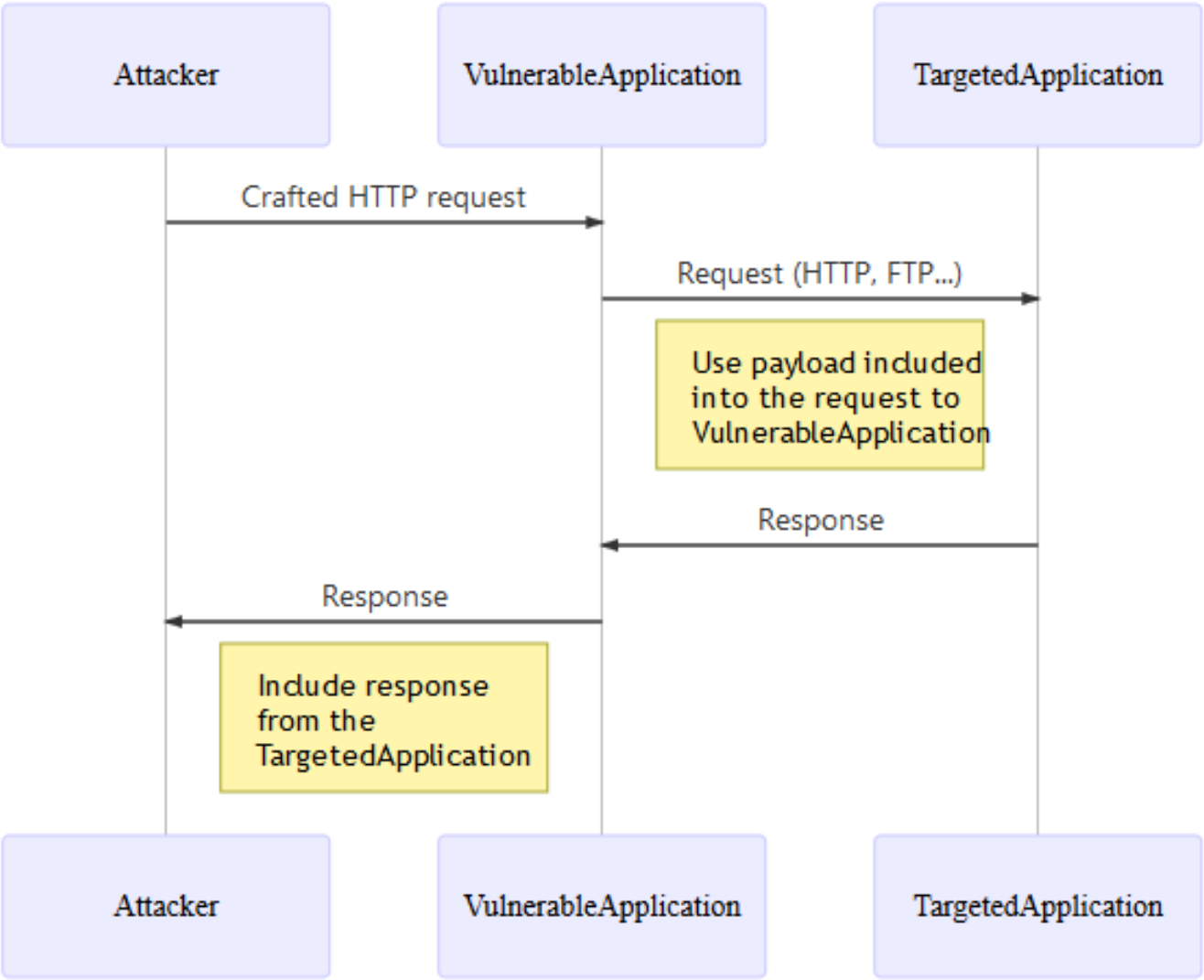


Implementation Steps for CSRF Protection – Receiver Channel

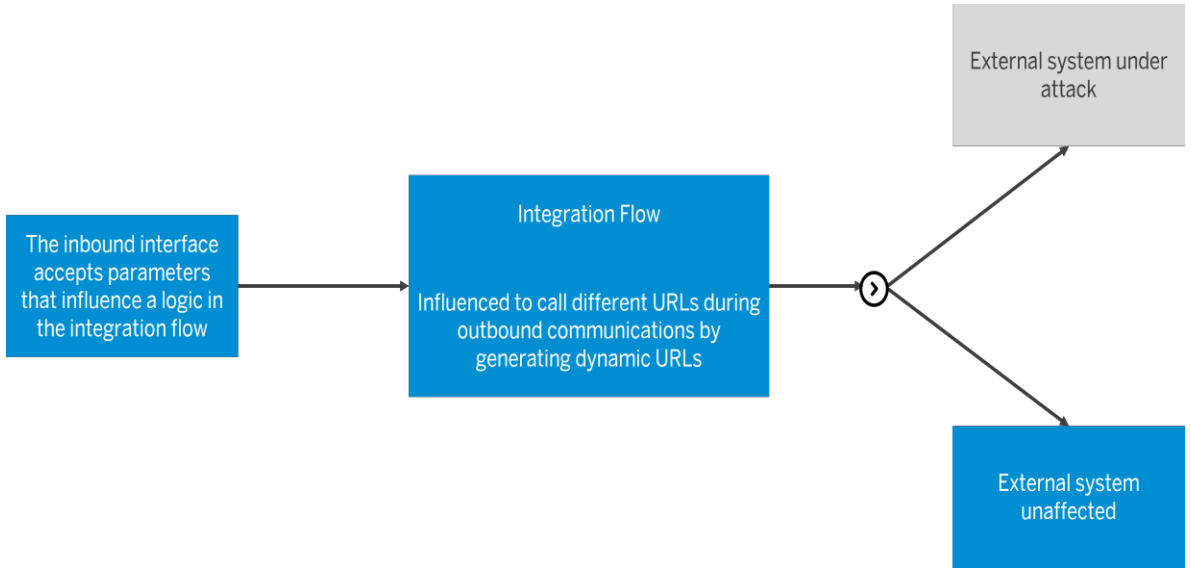




Server-Side Request Forgery (SSRF)



Guidance for SSRF



- **Validate all untrusted input** used to generate URLs for system-to-system interactions.
- **Allowlist headers** before passing them to network adapters (e.g., DB, cache, document services).
- **Restrict protocols** by allowing only secure ones like https://; reject others like file://, ldap://, gopher://, http://, and URLs with port numbers.
- **Add validation scripts** to check headers either after the sender channel or before the receiver channel.
- **Check against a configurable allowlist** of permitted IP addresses and domain names for destination access.
- **Avoid dynamic URLs** by using preconfigured IPs or destinations instead of parameters like CamelHttpUri.
- **Ensure URLs are on an allowlist** before using them.
- **Enforce strict authentication and authorization** for all data-exposing endpoints.
- **Design with caution**: avoid granting powerful egress access based on low-privilege inbound authorization.
- **Validate fetched data**: confirm the response body matches expectations; fail the flow if it doesn't.

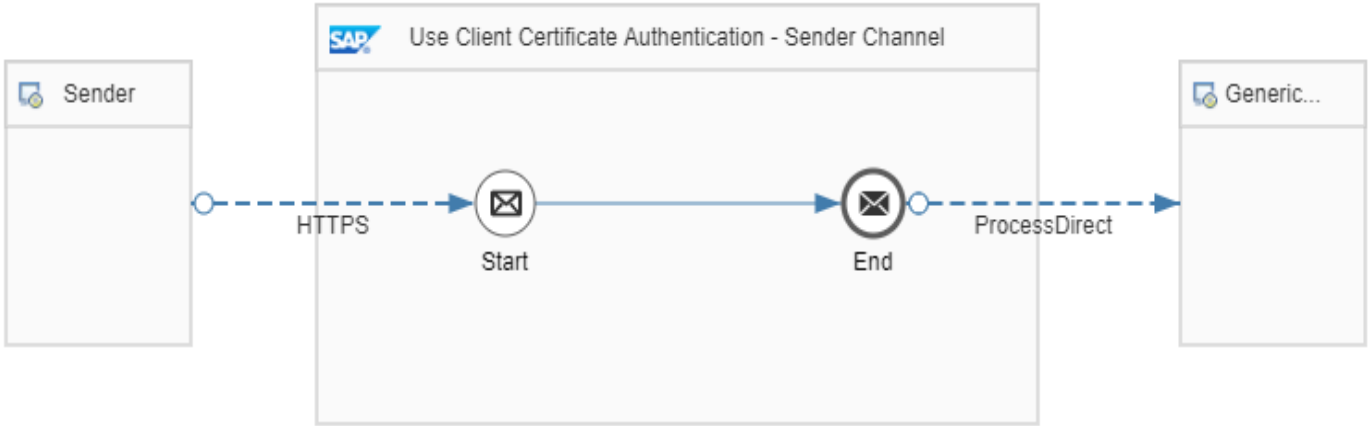
Secure Authentication Methods

Inbound Communication Authentication Options:

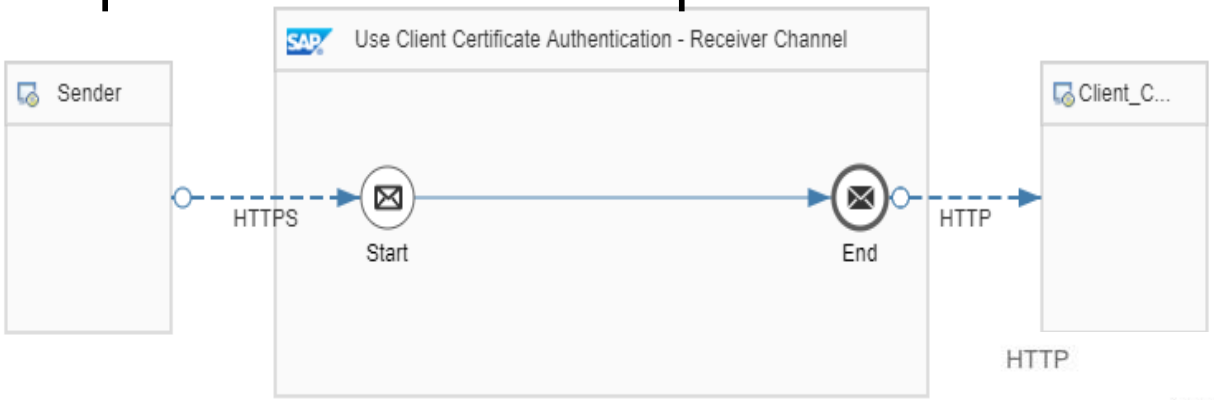
- **None (ODATA, AMQP, HTTP, SOAP, …)**
- **Basic**
  - Using the Identity Provider (IdP), which is SAP by default
  - via Service Instances
- **Principal Propagation**
- **SASL (AMQP)**
- **OAuth (Token Based: ODATA , AMQP)**
  - Client Credentials (via Service Instances)
- **Client Certificate**
  - via Service Instances
  - via Integration Flow

Please note: SAP does not recommend basic authentication (neither option) in productive scenarios.

Implementation Example with Client Certificate Authentication(Part 1)



Implementation Example with Client Certificate Authentication(Part 2)

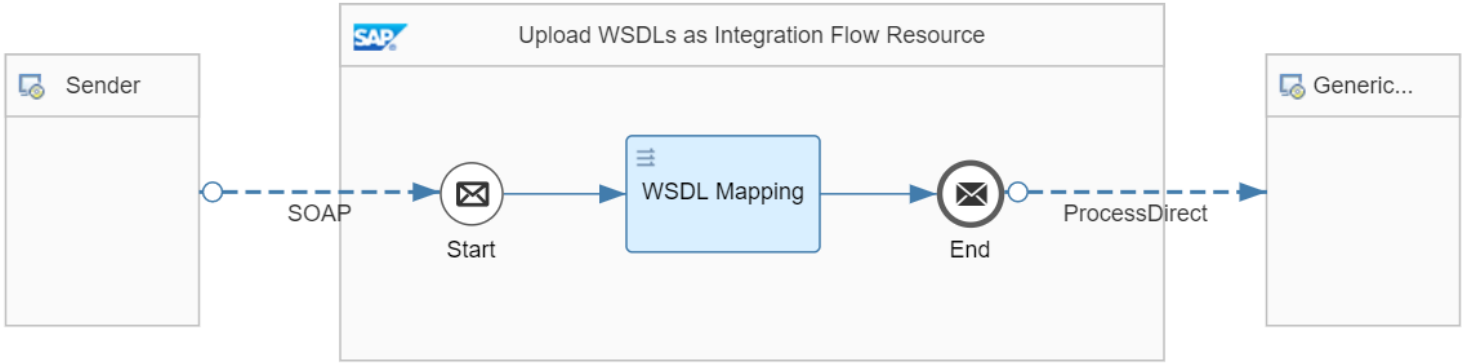


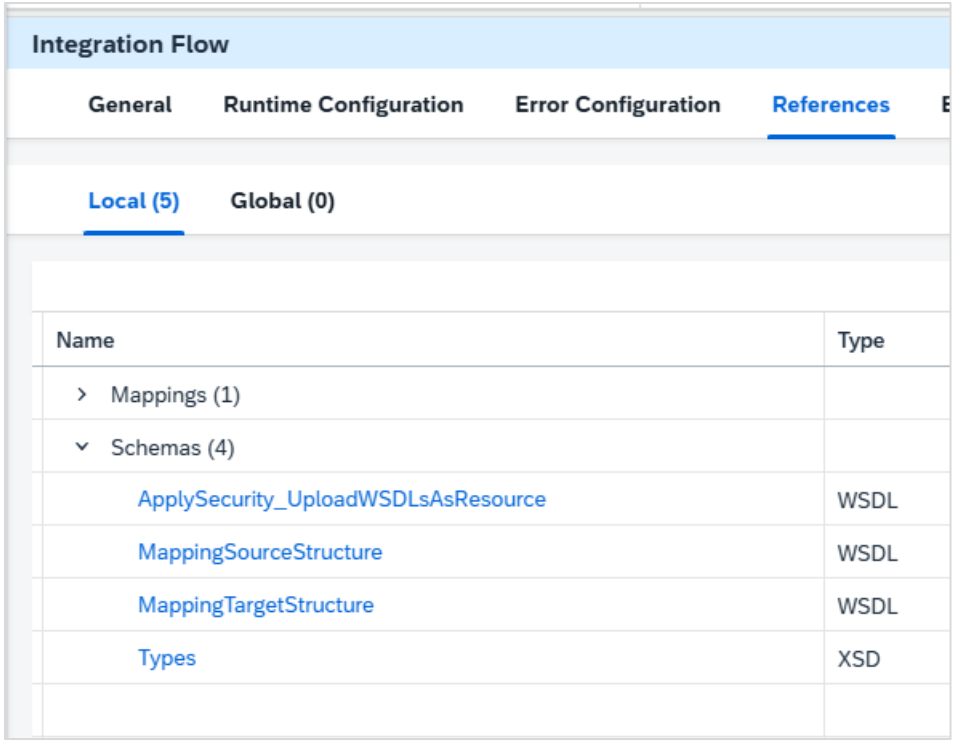
General | Connection

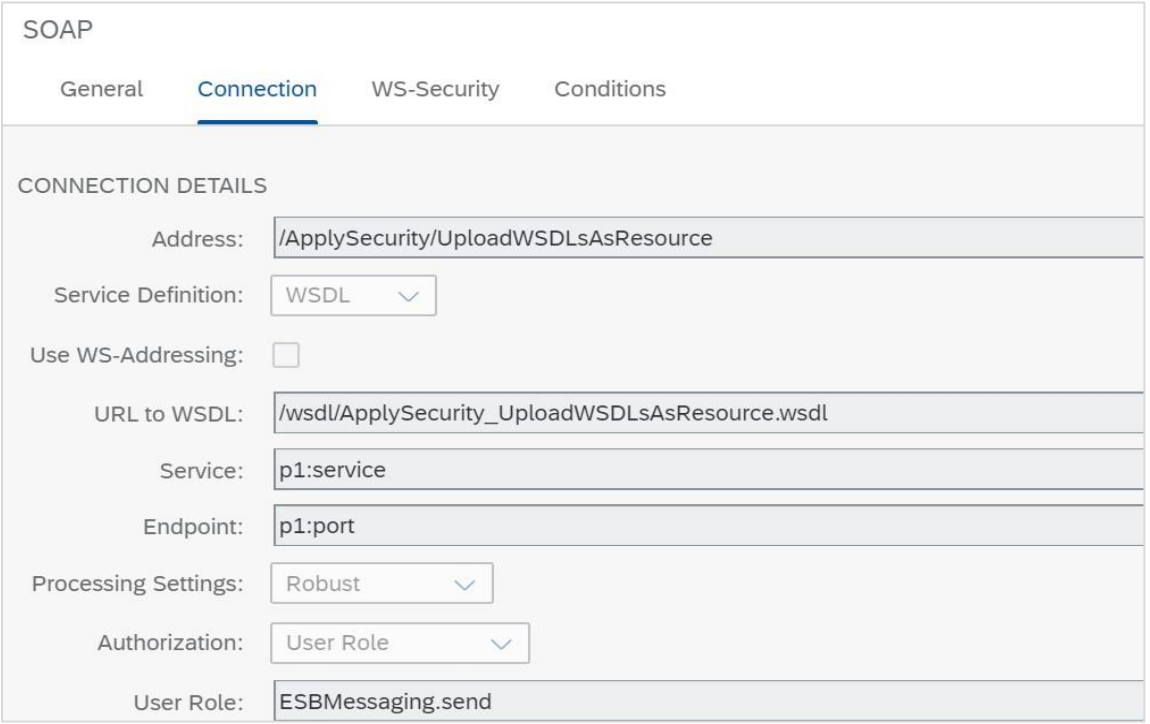
CONNECTION DETAILS

Address:	https://${header.target_host}/http/ApplySecurity/UseClientCertAuthInternal
Query:	
Proxy Type:	Internet
Method:	GET
Send Body	
Authentication:	Client Certificate
Private Key Alias:	${header.keystore_alias}
Timeout (in ms):	60000

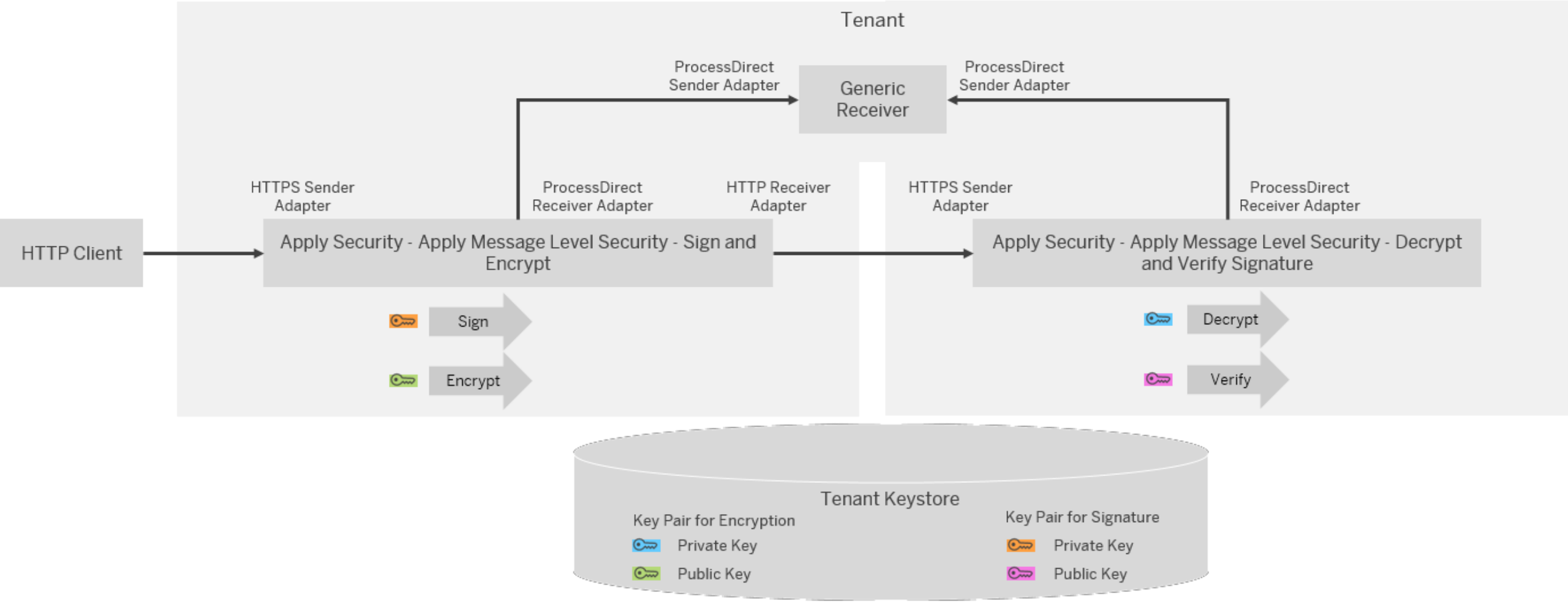
Upload WSDLs as Integration Flow Resource



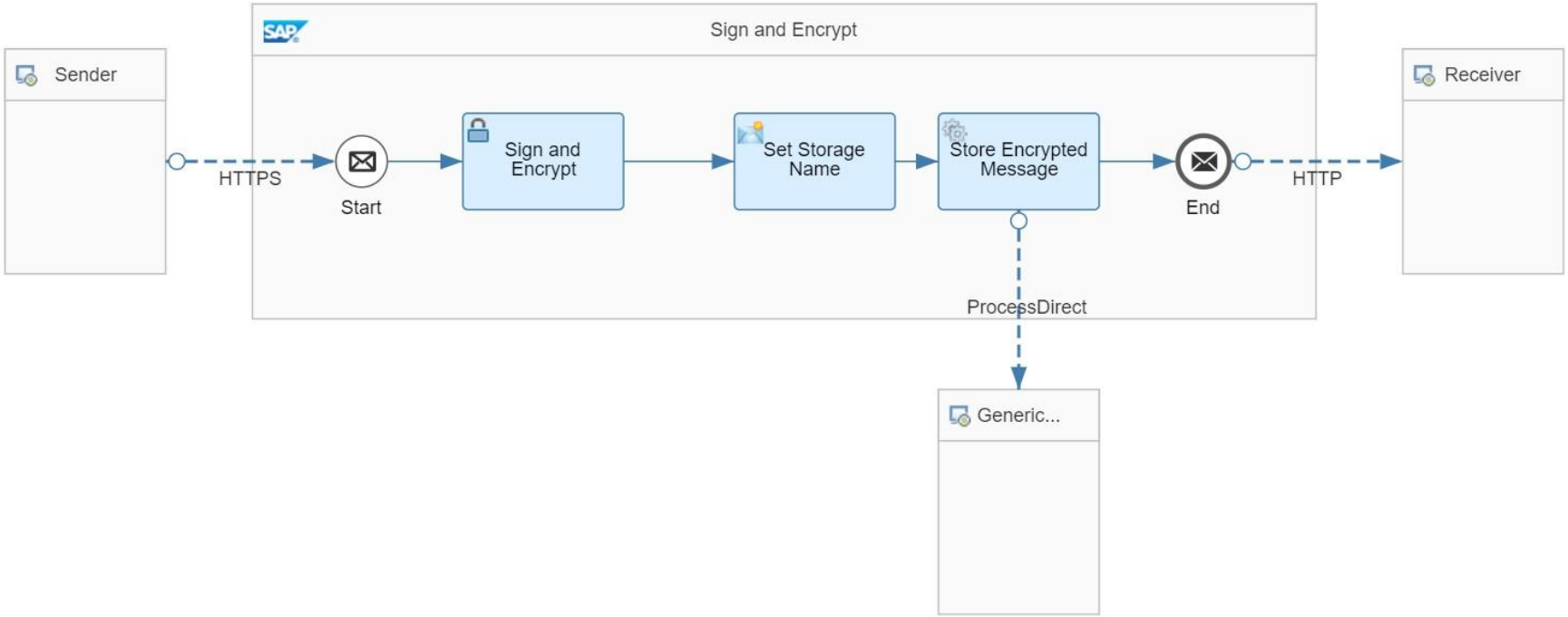


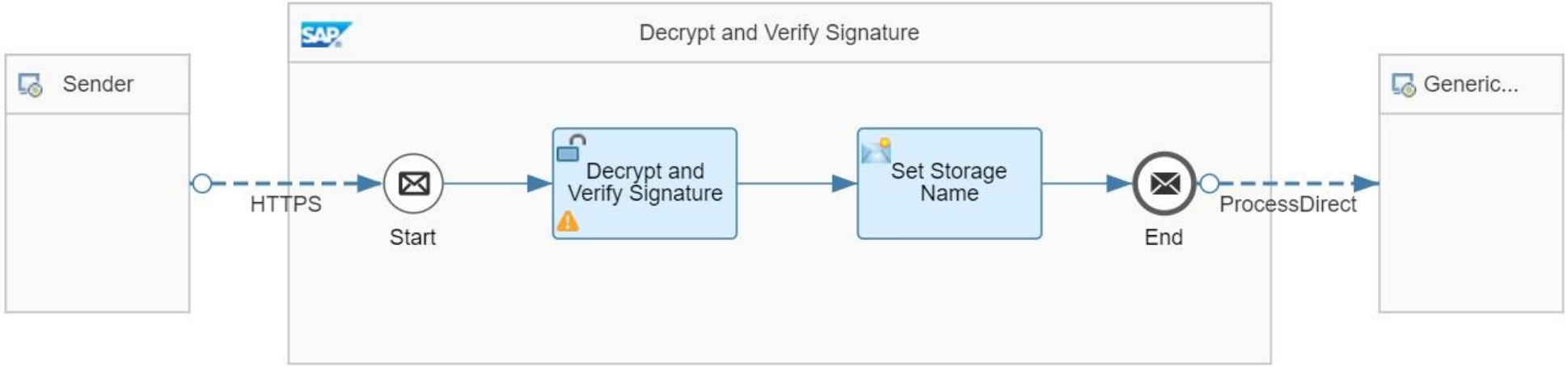


Message-Level Security

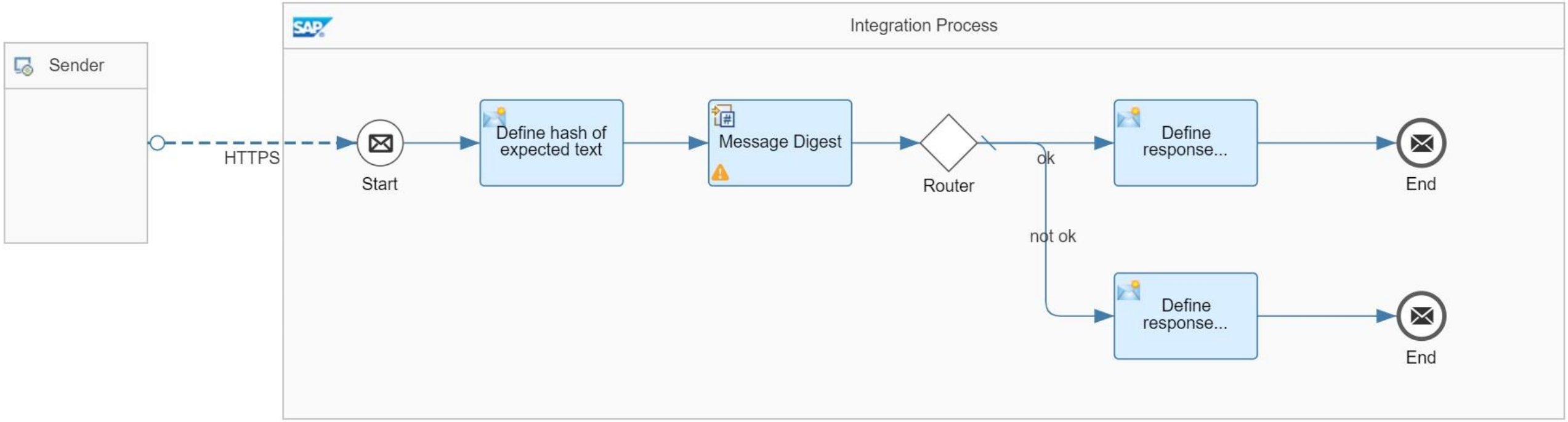


Implementation Example with Message Level Security

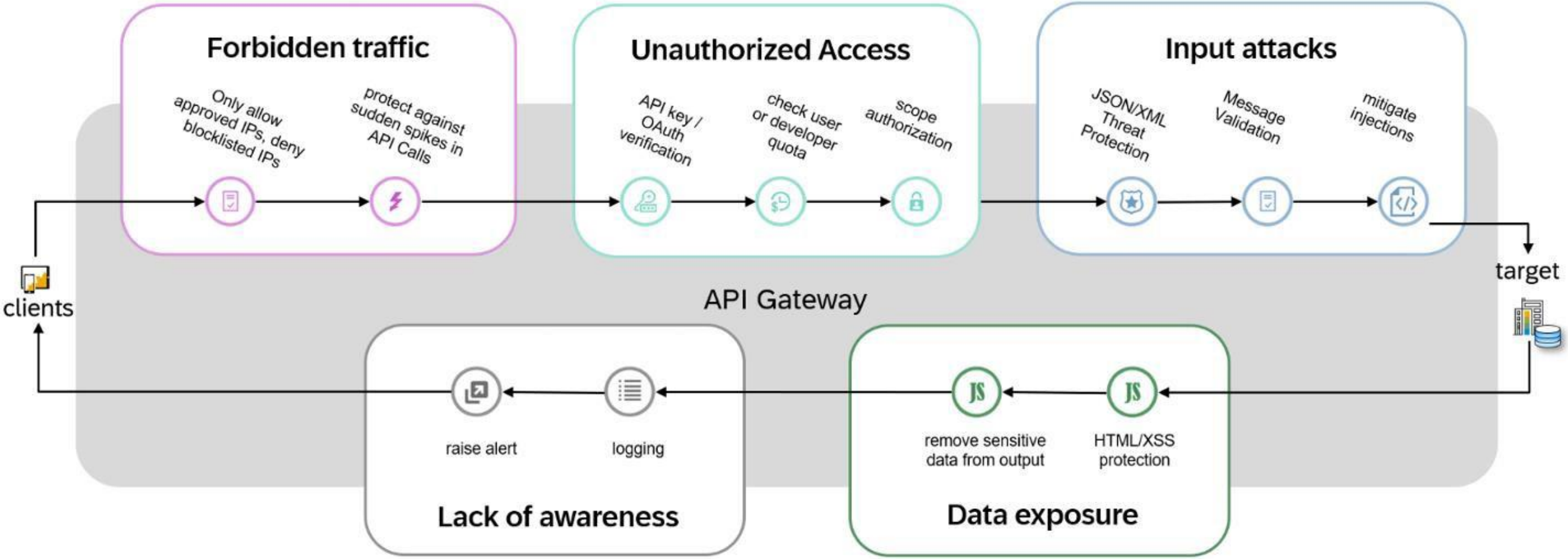




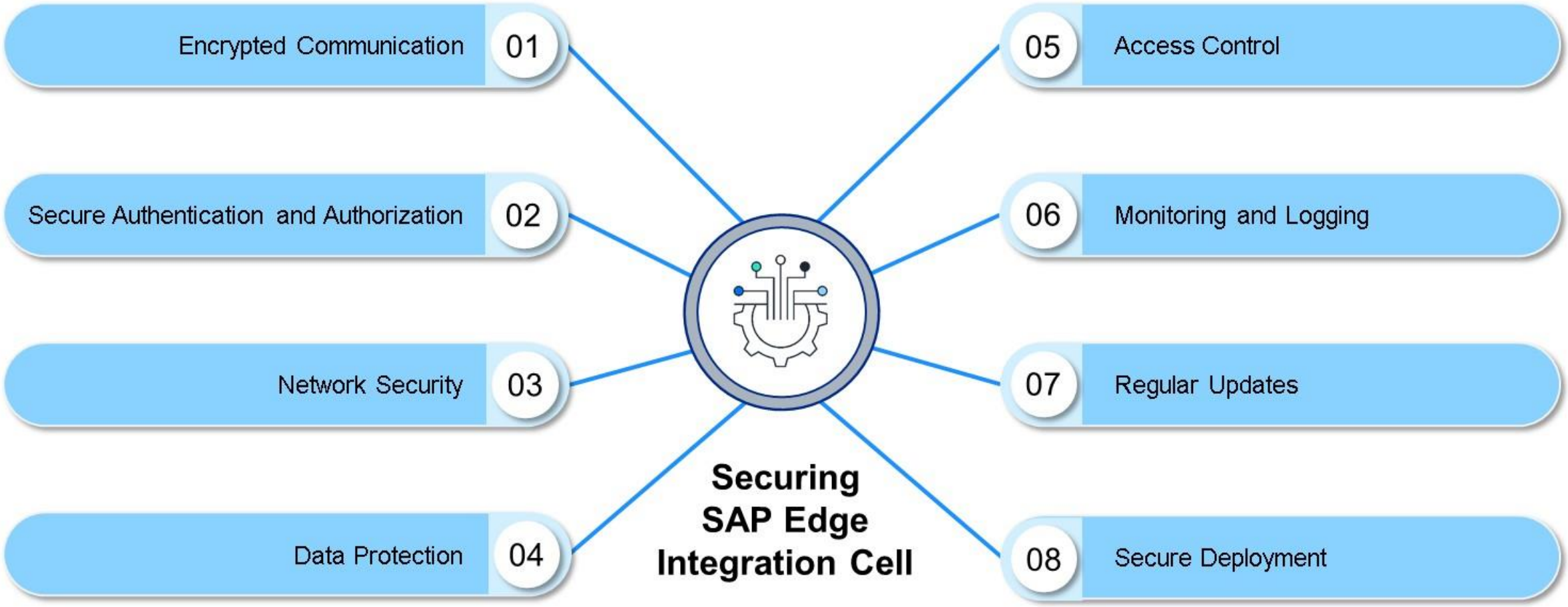
Protect Data Integrity



API Security



Securing SAP Edge Integration Cell



Questions

Sign up SAP BTP

Sign up for a PAYG account* and start today with our free tier service plans!

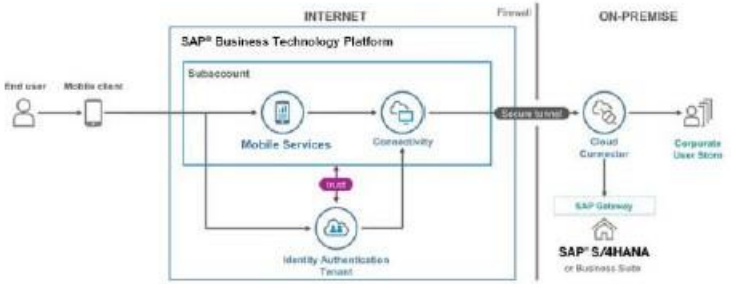
Need ideas? Check out the “Free Tier” missions on SAP Discovery Center. Start developing – check out missions

Free tier services → Free tier missions → SAP Community for SAP BTP

POWERING SAP BTP

- Access free-tier service plans
- Start developing – check out missions
- Go productive with a simple upgrade
- Get support from the entire SAP community

Empower field technicians by building native apps using SDKs



** Existing CPEA & PAYG customers have access to free tier service plans*

Thank you.

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Appendix

SAP Business Technology Platform for RISE

Consume innovation faster

Make data-driven decisions with confidence using reliable master data

SAP Master Data Governance

Keep the core clean to accelerate innovation by extending SAP S/4HANA Cloud, private edition

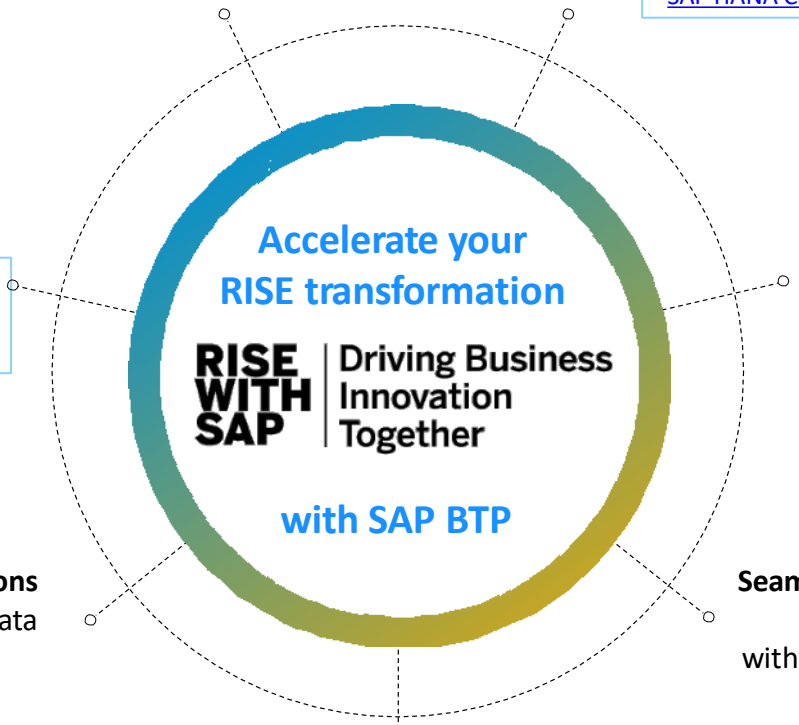
SAP Build
SAP HANA Cloud

Accelerate data migration with best practices and proven solutions

SAP Advanced Data Migration by Syniti SAP Enterprise Continuous Testing by Tricentis
SAP Data Services

Automate SAP S/4HANA Cloud, private edition processes using prebuilt automations

SAP Build Process Automation



Power strategic decisions with comprehensive data warehousing solutions

SAP Datasphere

Seamlessly integrate SAP S/4HANA Cloud, private edition with applications in your landscape

SAP Integration Suite

Achieve 360° insights with a single connected analytics platform

SAP Analytics Cloud

Infuse AI into every aspect of the business

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Connect Business Processes

Key Product Information

SAP Integration Suite

https://www.sap.com/products/technology- platform/integration-suite.html

Integration Space

L1 - SAP Integration Suite

L2 – SAP Integration Suite

Guidelines : https://community.sap.com/t5/technology- blogs-by-sap/integration-developer-s-corner/ba- p/13349213

SAP BTP drives Enterprise Transformation

Infuse AI into every aspect of the business

Accelerate your move to the cloud

Connect business processes

NOW

ECC or S/4HANA

Customers building and planning their SAP S/4HANA Cloud journey

Move, cleanse and maintain clean data & test (SAP MDG, Solution Extension)

Migrate SAP HANA data (SAP HANA Cloud)

Analyze Process Insights (SAP Signavio)

Integrate business processes (SAP Integration Suite)

Extend and automate

Create custom Business Sites (SAP Build Work Zone)

Continuously automate processes (SAP Build Process Automation)

Easily extend SAP S/4HANA (SAP Build Apps)

Conduct custom code analysis (SAP BTP)

Plan and analyze

Simplify and access data to all business users (SAP Datasphere)

Plan and Forecast (SAP Analytics Cloud for planning

Deliver real-time insights from all data (SAP Analytics Cloud)

202X

Intelligent, Sustainable Enterprise